

JKBOSE

2026 CLASS 11TH

MOTION IN STRAIGHT LINE

LECTURE-3

PHYSICS

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Topics To Be Covered

- 1 Revision
- 2 Acceleration ✓
- 3 Positive Acceleration ✓✓
- 4 Negative Acceleration ✓
- 5 Zero Acceleration ✓

Graph

Recap From 1st Lec.

Motion

- move one place to another place with respect to

Rest

Donot move from one place to another place with respect to

Reference point
(Motion is Relative)

Numericals

$$1 \frac{\text{km}}{\text{hr}} = \frac{5}{18} \text{ m/s}$$

distance

- Total path
- metre
- Scalar
- Positive

Speed

- $\text{Speed} = \frac{\text{distance}}{\text{time}}$
- Scalar
- m/s
- +ive

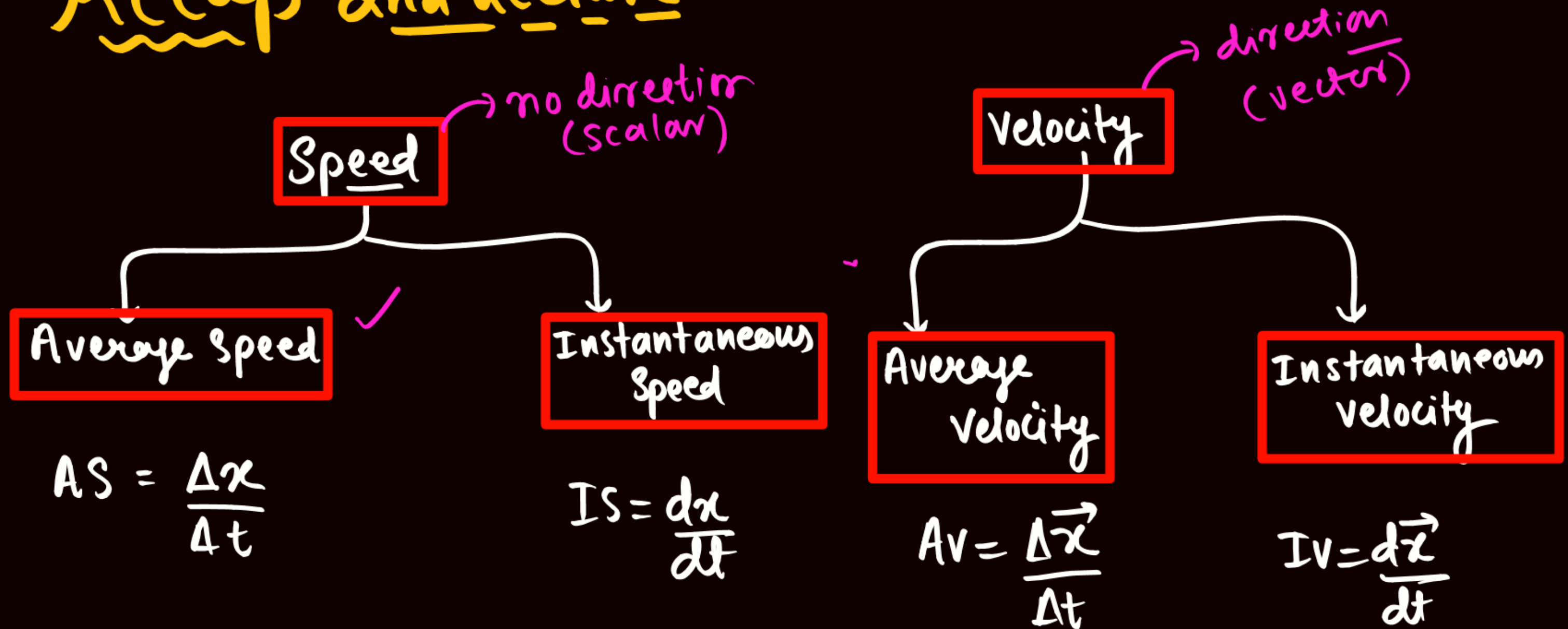
displacement

- Shortest path
- metre
- vector
- + | - | zero

Velocity

- $\text{Velocity} = \frac{\text{displacement}}{\text{time}}$
- Vector
- m/s
- +ive | -ive | 0

Recap and lecture



Topic - ①

Lecture no. 3

($\vec{a}$) Acceleration :- Rate of change of $\vec{v}$ velocity

Vector

$$a = \frac{\Delta v}{t}$$

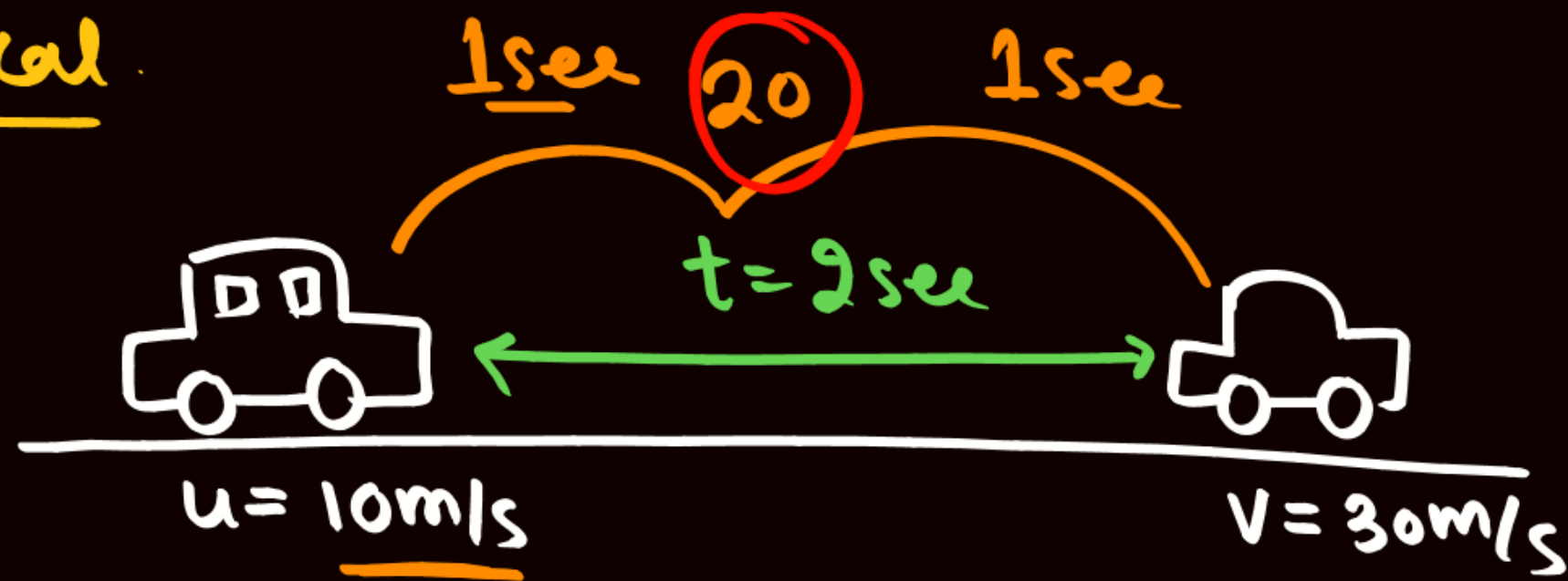
$$\# a = \frac{\Delta v}{t} = \frac{m/s}{s} = \frac{m}{s^2}$$

$$a = \frac{v_f - v_i}{t}$$

$$\vec{a} = \frac{\vec{v} - \vec{u}}{t}$$

Numerical

Q①



$$a = ?$$

So,

$$a = \frac{v - u}{t}$$

$$a = \frac{30 - 10}{2} = \frac{20}{2} = +10 \text{ m/s}^2 \text{ (JKBOSE)}$$

m/s²

Note

Acceleration

✓ Rate of change of velocity

$$\vec{a} = \frac{\Delta \vec{v}}{t}$$

• Vector ✓

• unit

$$a = \frac{v_f - v_i}{t}$$

$$a = \frac{v - u}{t}$$

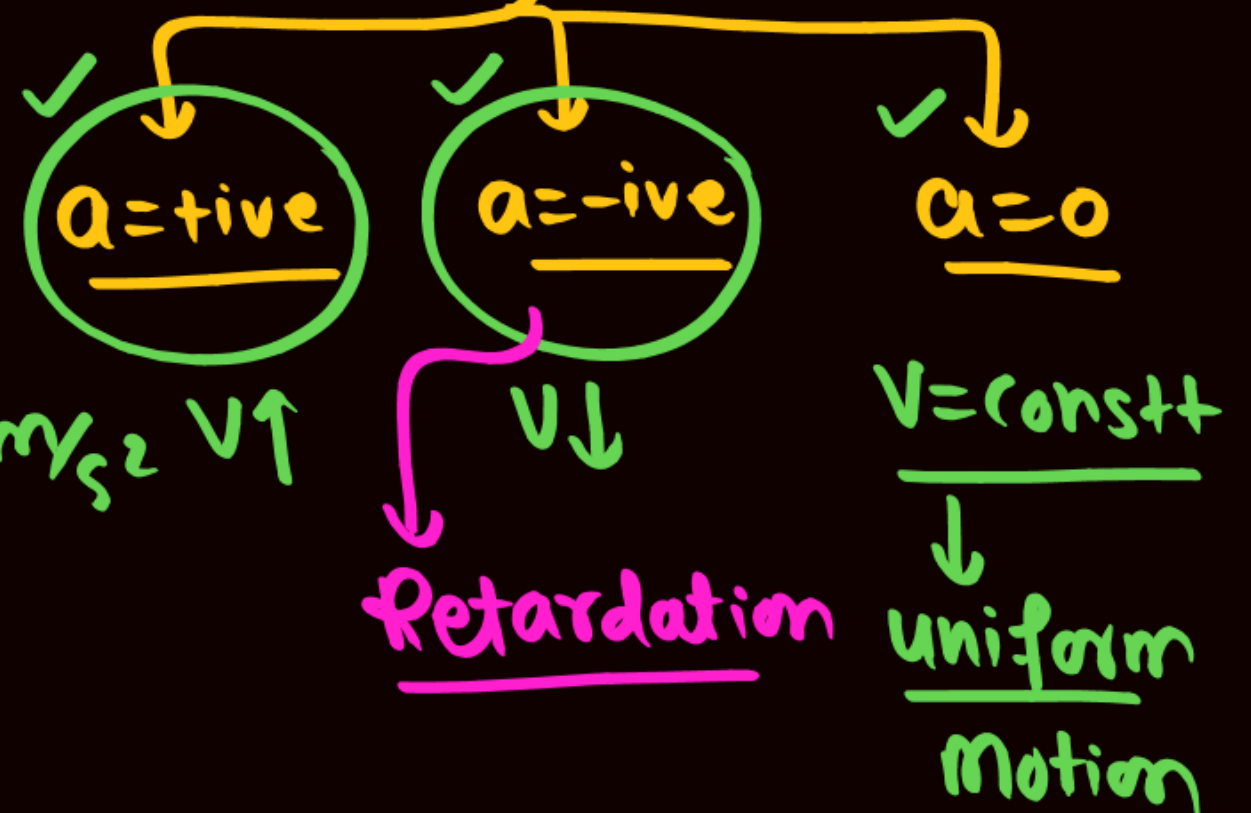
$$at = v - u$$

$$u + at = v$$

✓ $v = u + at$

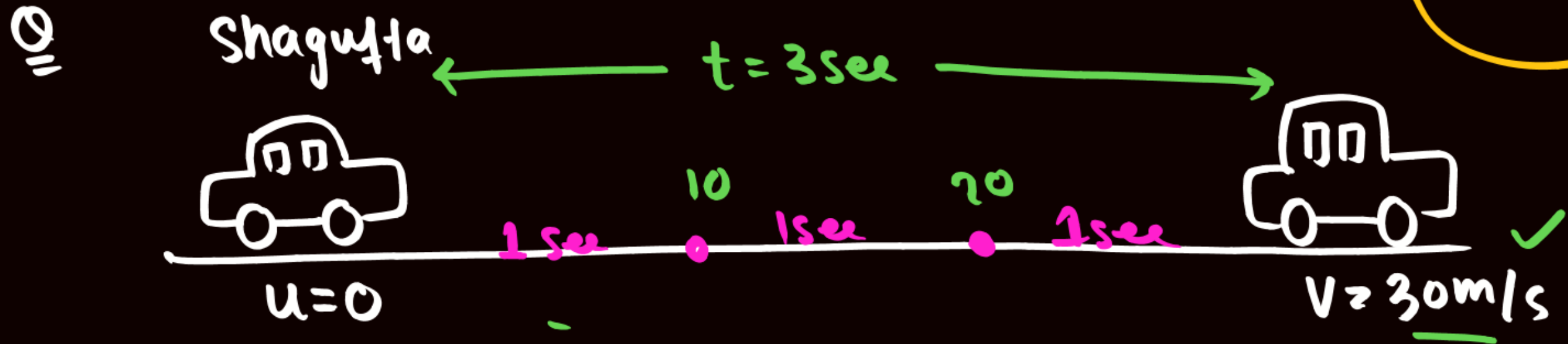
$$a = \frac{\Delta v}{t} = \frac{m/s}{s} = \frac{m}{s \times s} = m/s^2 \quad v \uparrow$$

Numerical



Numerical - (2) Positive acceleration

$$a = ?$$



So

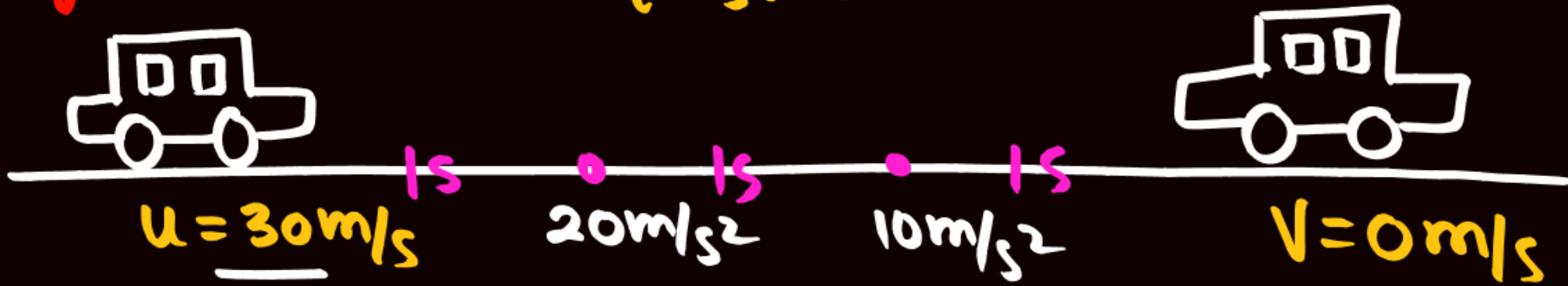
$$a = \frac{V - u}{t}$$

$$a = \frac{30 - 0}{3} = \frac{30}{3} = +10 \text{ m/s}^2$$

(non-uniform motion)

Numerical ③ (Break)
(Negative Acceleration)
 $t = 3 \text{ sec}$

$a = ?$



Solⁿ

$$a = \frac{v - u}{t}$$

$$a = \frac{0 - 30}{3} = -\frac{30}{3} = -10 \text{ m/s}^2$$

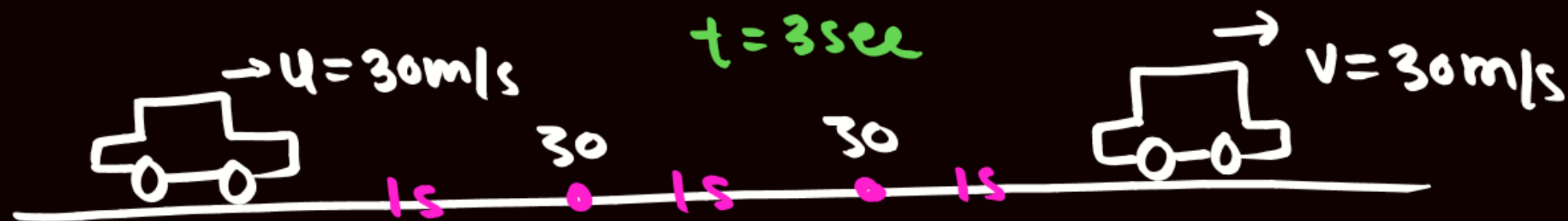
Retardation ✓

- Velocity ↓↓↓
- non-uniform motion ✓

Numerical - (4)

Zero acceleration

$$a = ?$$



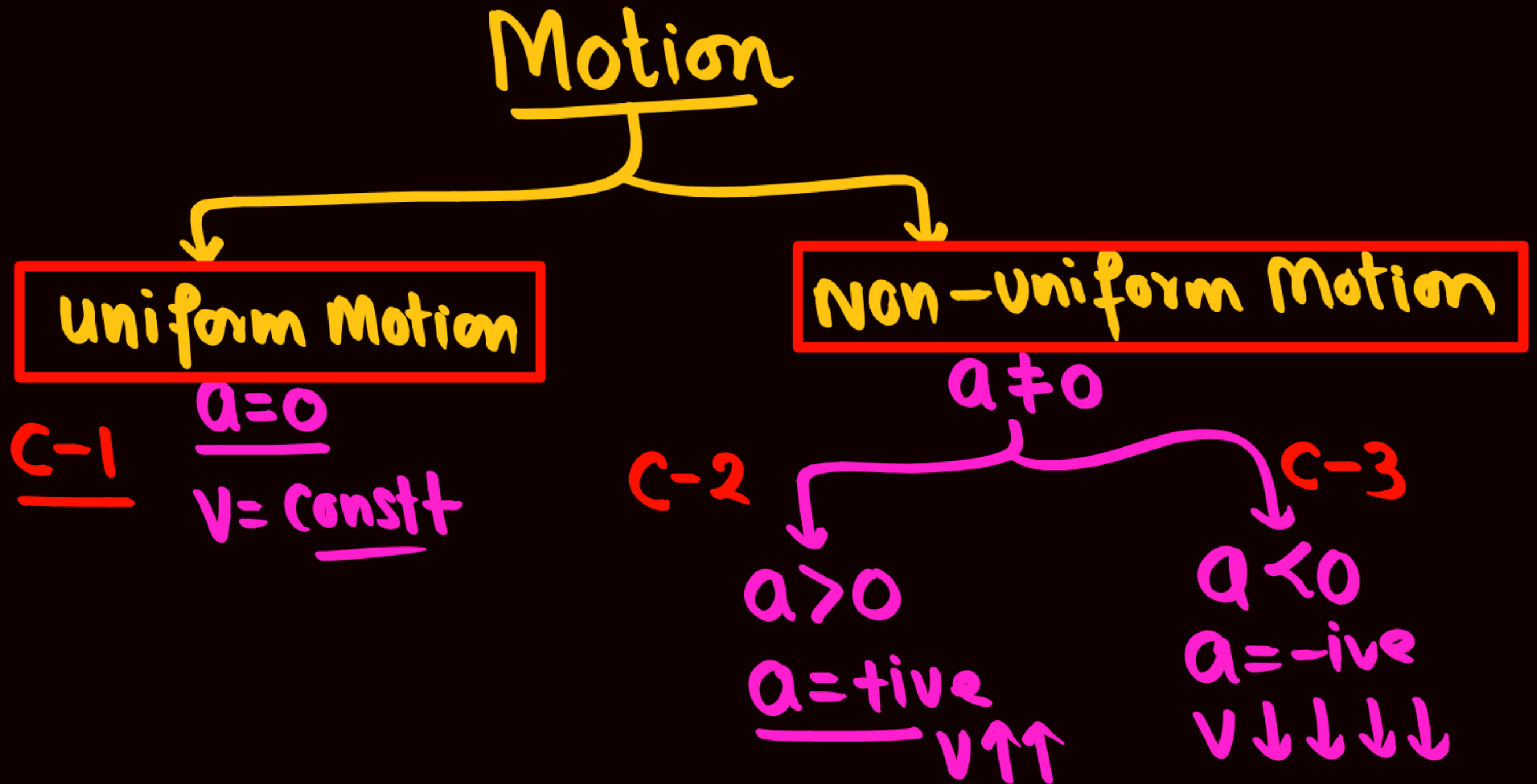
Sol

$$a = \frac{v - u}{t}$$

Uniform Motion

$$= \frac{30 - 30}{3} = \frac{0}{3} = 0 \text{ m/s}^2$$

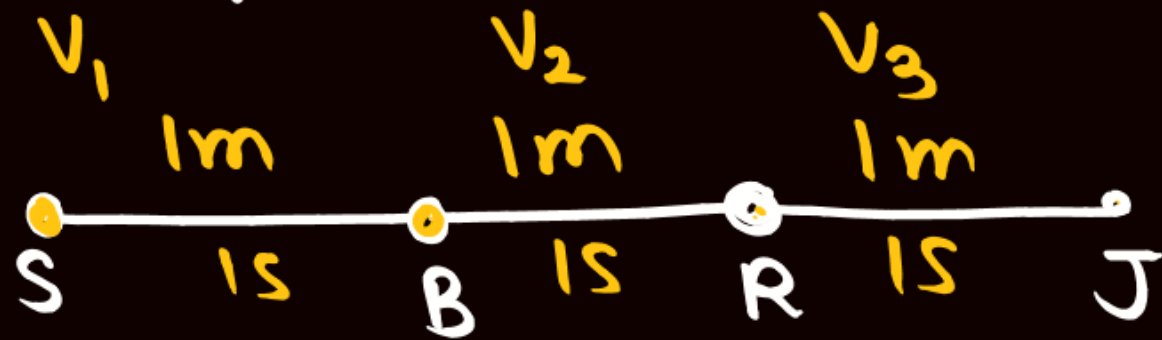
Topic 2nd (Types of motion) $\rightarrow$ Graph $\rightarrow$ x vs t ✓
 $\rightarrow$ v vs t ✓
 $\rightarrow$ a vs t ✓



C-1

Uniform Motion

- When Body cover equal displacement in equal interval of time



- Instantaneous velocity

$$V_1 = \frac{1m}{1s} = 1m/s \quad \checkmark$$
$$V_2 = \frac{1m}{1s} = 1m/s \quad \checkmark$$
$$V_3 = \frac{1m}{1s} = 1m/s$$

- Average Velocity

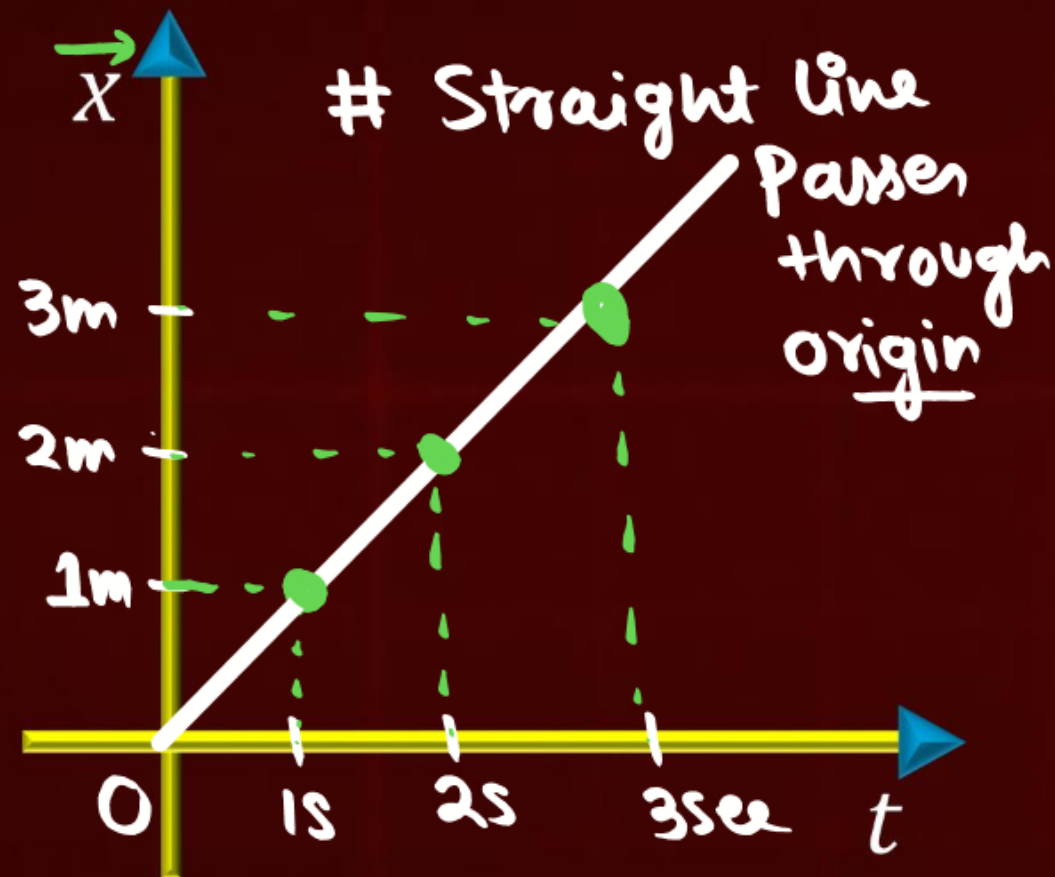
$$V_a = \frac{\sum d}{\sum t} = \frac{3m}{3s} = 1m/s$$

$$\# V_a = V_I \quad \checkmark \quad \text{Neet-2017}$$

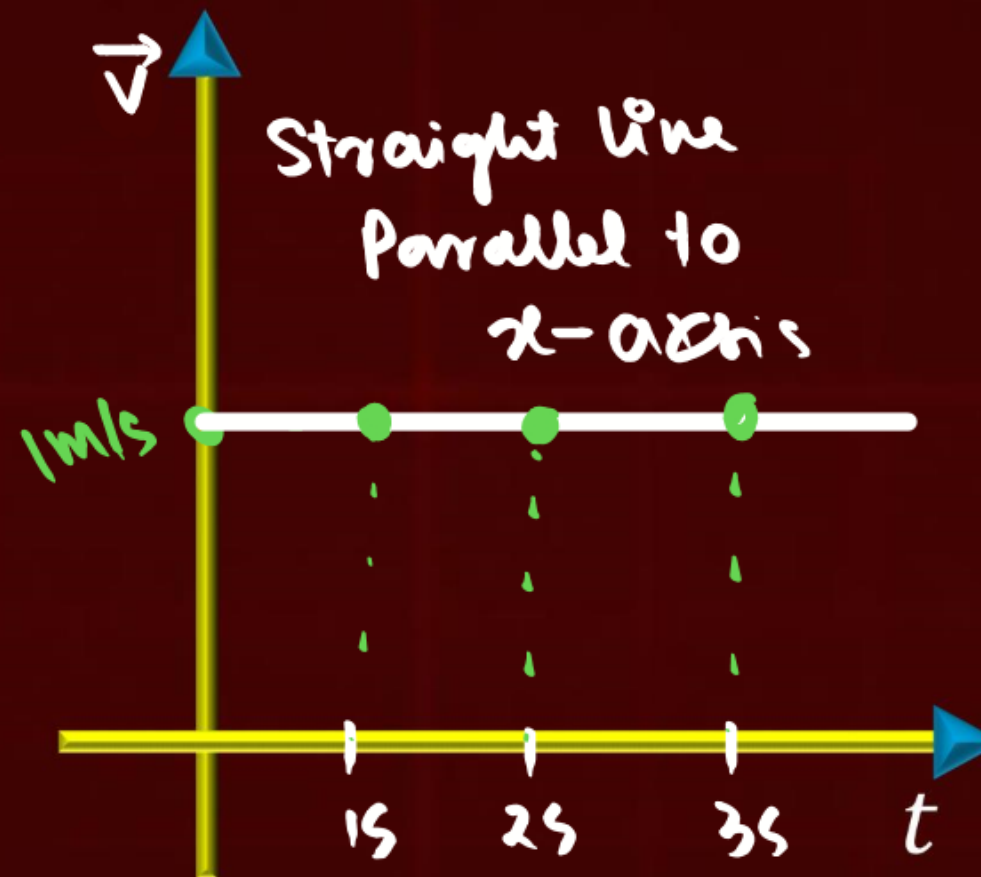
$$\# a = \frac{V_f - V_i}{t} = \frac{V - U}{t} = \frac{1 - 1}{1} = \frac{0}{1} = 0m/s^2$$

Graphs

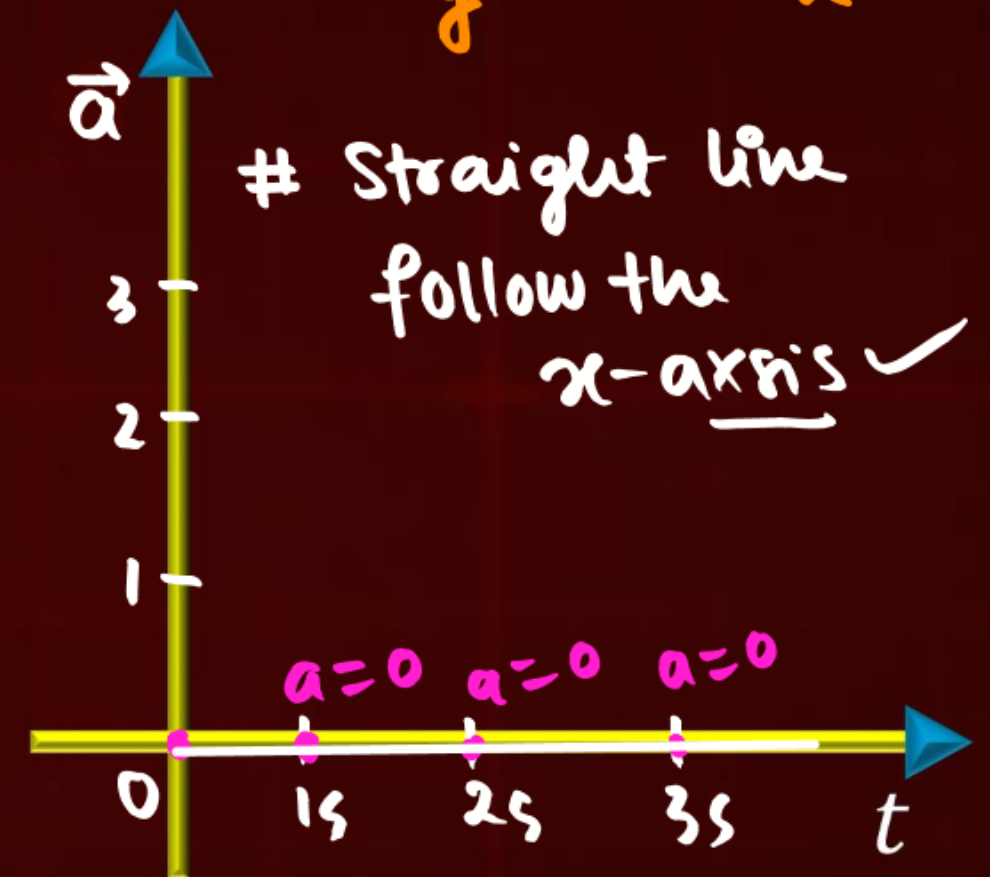
① displacement $\vec{x}$ time t graph
y x



② velocity - time
y x



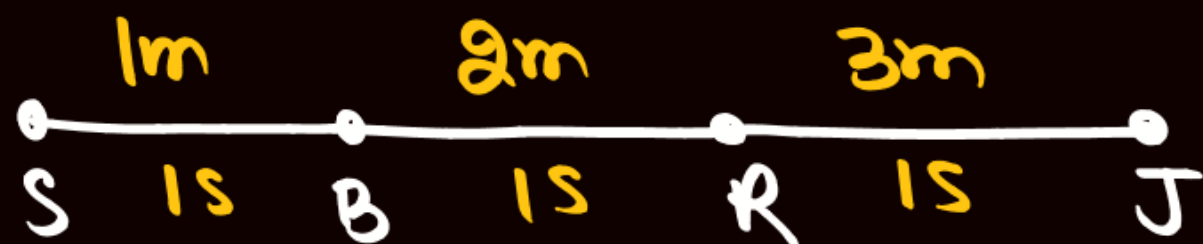
acceleration time
y x



C-2

Non-uniform motion ($v \uparrow \uparrow$)

Body cover un-equal displacement in equal interval of time



Instantaneous velocity

1s
1s

$$V_1 = \frac{1m}{1s} = 1m/s$$
$$V_2 = \frac{2m}{1s} = 2m/s$$
$$V_3 = \frac{3m}{1s} = 3m/s$$

$V_1 < V_2 < V_3$

Velocity average.

$$V_a = \frac{Td}{Tt} = \frac{1+2+3}{1+1+1} = \frac{6}{3} = 2m/s$$

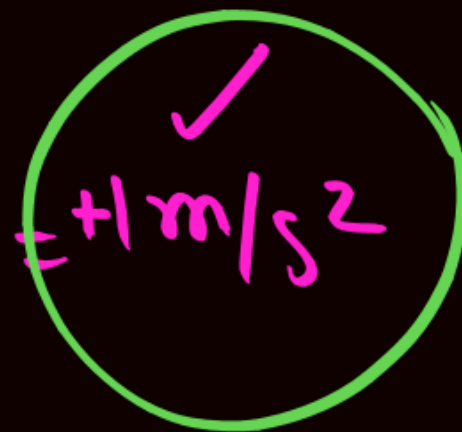
$V_a \neq V_I$

Acceleration

$$a = \frac{V_f - V_i}{t}$$

$$= \frac{2 - 1}{1} = 1 = +1m/s^2$$

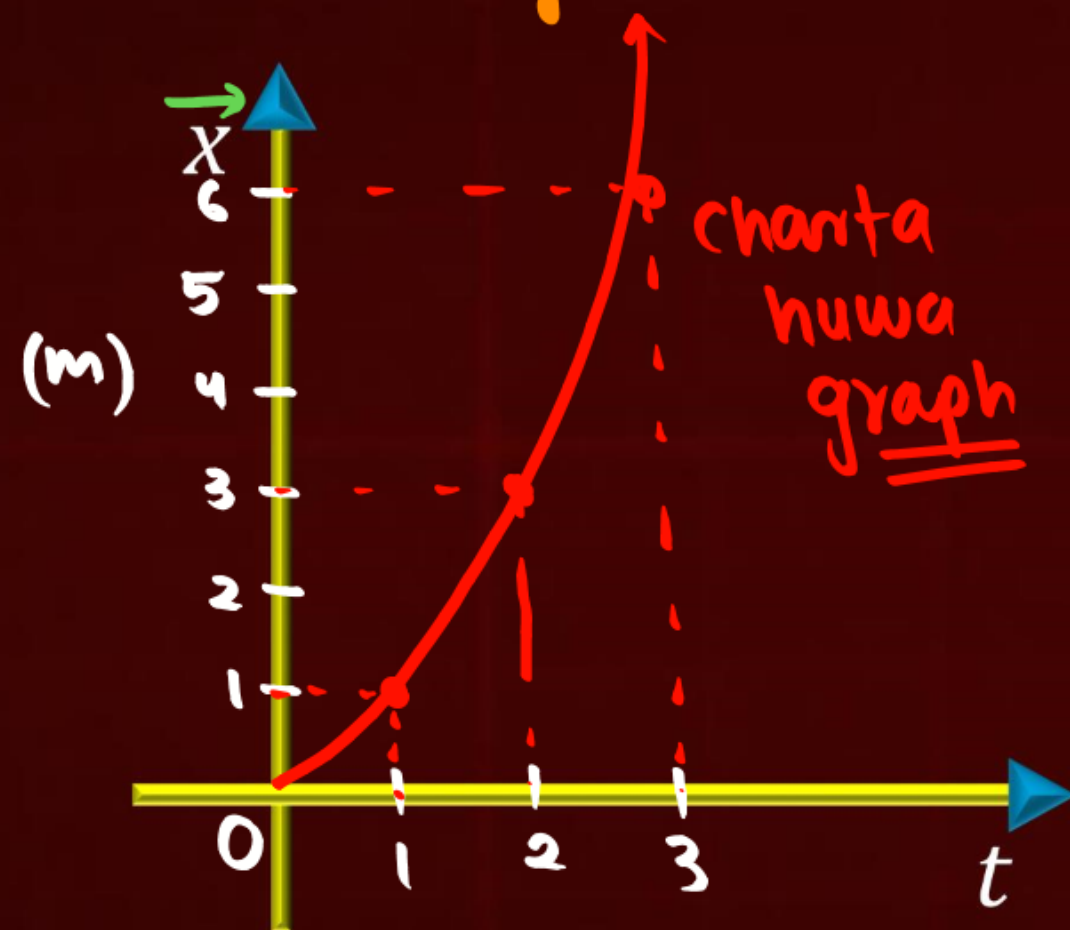
$v \uparrow \uparrow \uparrow \uparrow$



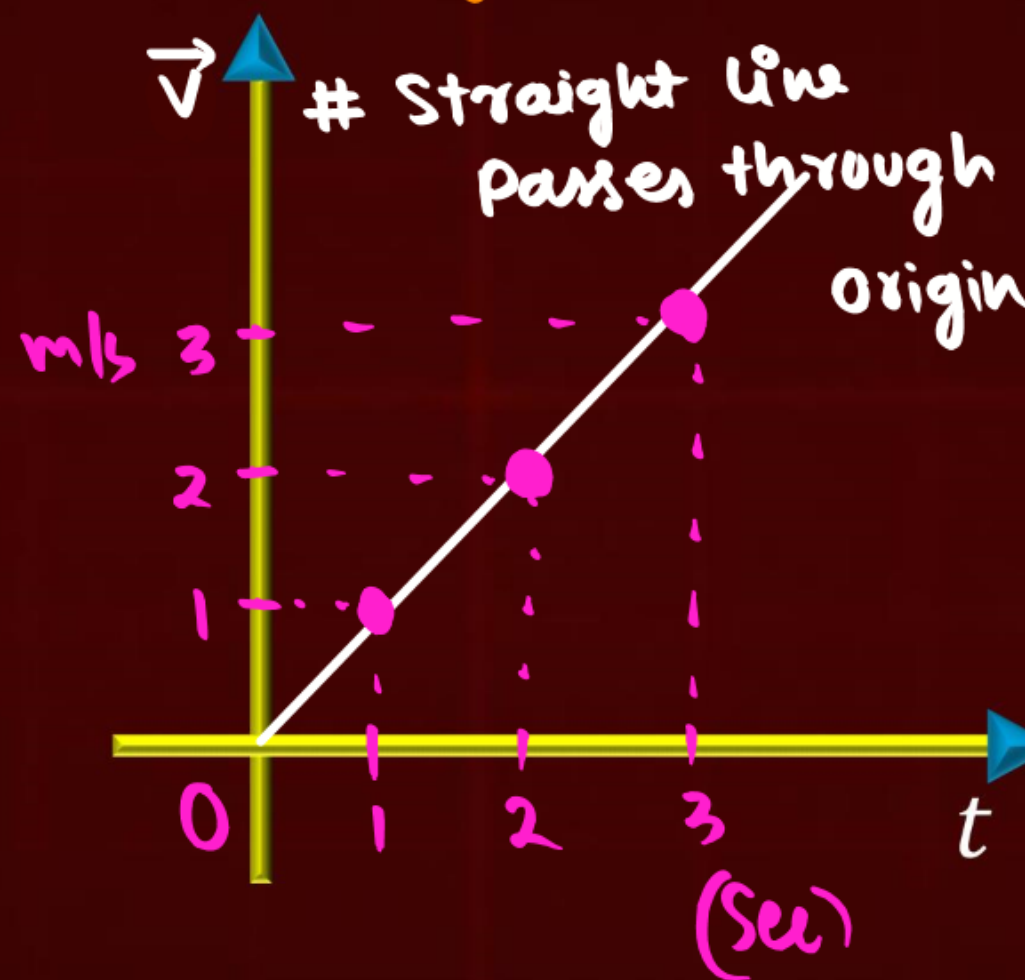
Graphs



① displacement $\vec{x}$ time t graph
y x



② velocity - time
y x



acceleration time
y x



C-3 (non-uniform Motion)

↓↓↓↓
(Slow)
(Break)

$$\# V_a = \frac{T_d}{T_t} = \frac{1m+1m+1m}{1+2+3} = \frac{3}{6} = \frac{1}{2} = 0.5m/s$$

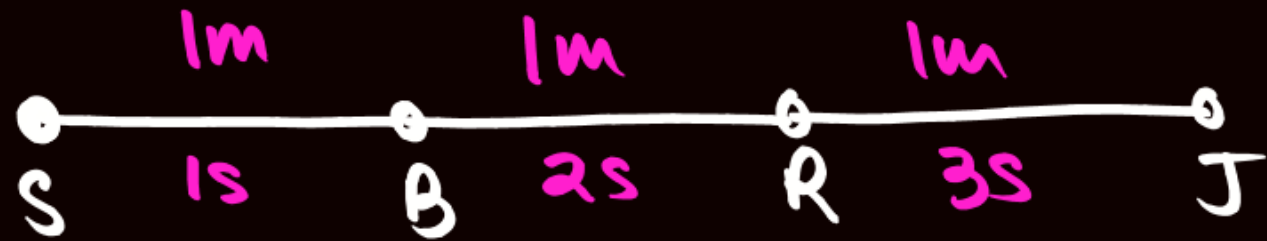
When Body cover equal displacement in un-equal interval of time

$$\# V_I \neq V_a$$

$$\# a = \frac{V-u}{t}$$

$$a = \frac{0.5-1}{1} = -0.5 = -0.5m/s^2$$

Retardation
(Break) ✓



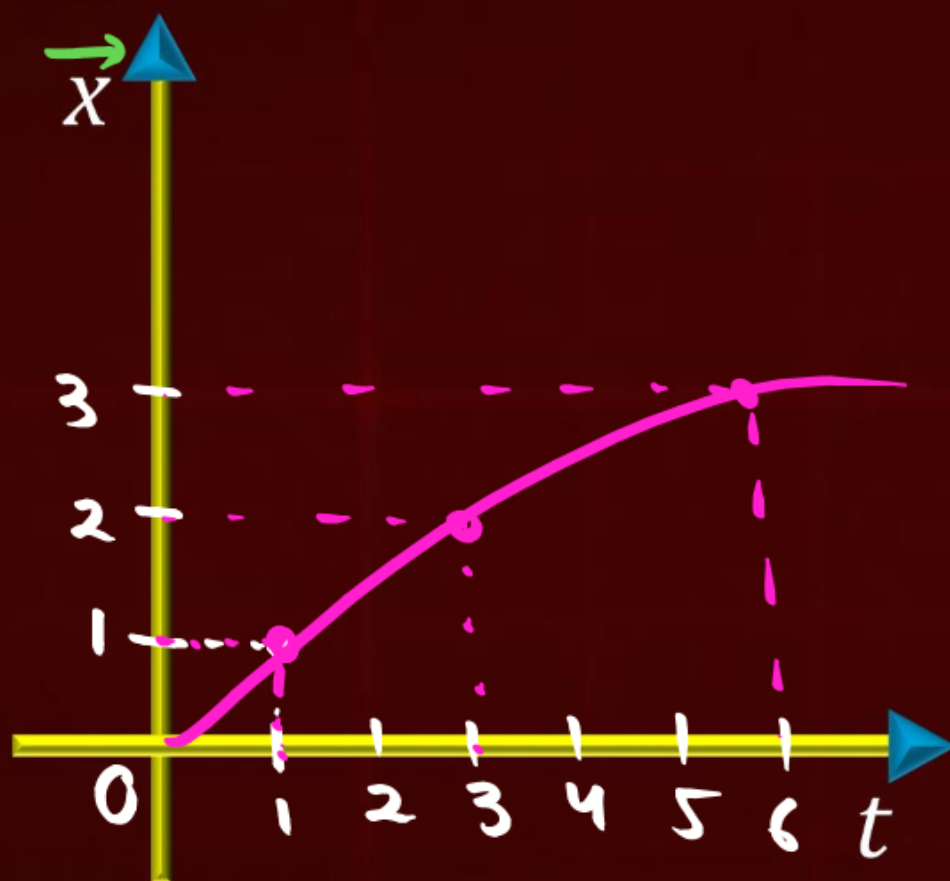
Instantaneous Velocity

$$V_1 = \frac{1m}{1s} = 1m/s$$
$$V_2 = \frac{1m}{2s} = \frac{1}{2}m/s = 0.5m/s$$
$$V_3 = \frac{1m}{3s} = 0.33m/s$$
$$V_1 > V_2 > V_3$$

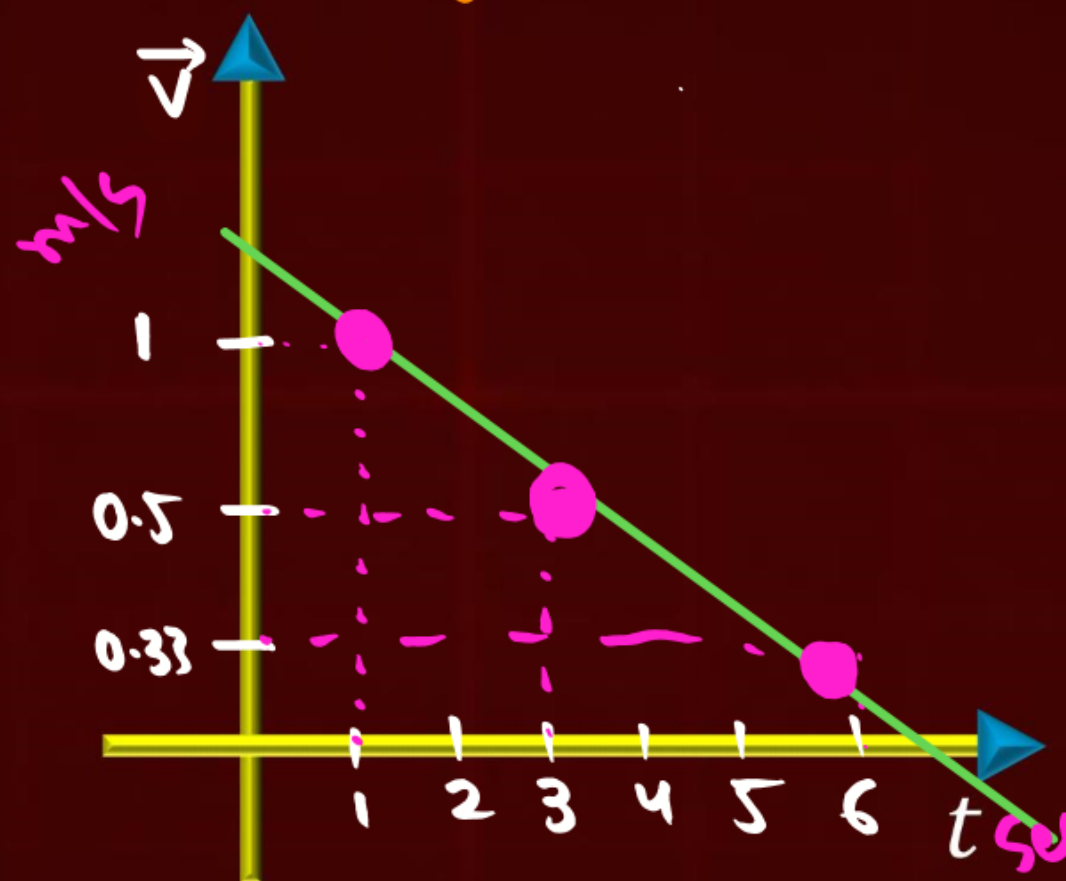
Graphs



① displacement $\vec{x}$ time t graph
y x



② velocity - time
y x



$a = -0.5 \text{ m/s}^2$
acceleration time
y x



SUMMARY

96

✓ acc

✓ acc +ive

✓ acc -ive

✓ $a = 0$

✓ u-motion $a = 0$

✓ non-u-motion

→ $a > 0$

→ $a < 0$

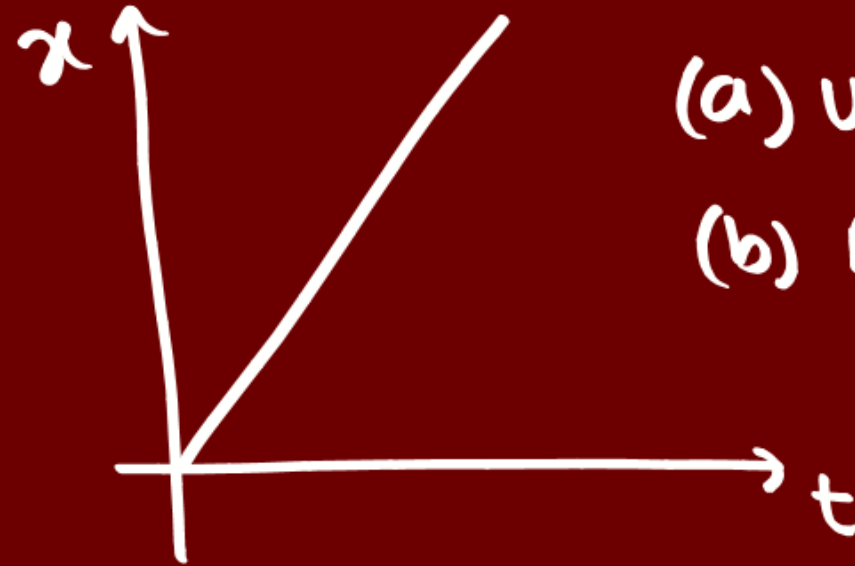
Slope ✓

Area ✓

HOMWORK



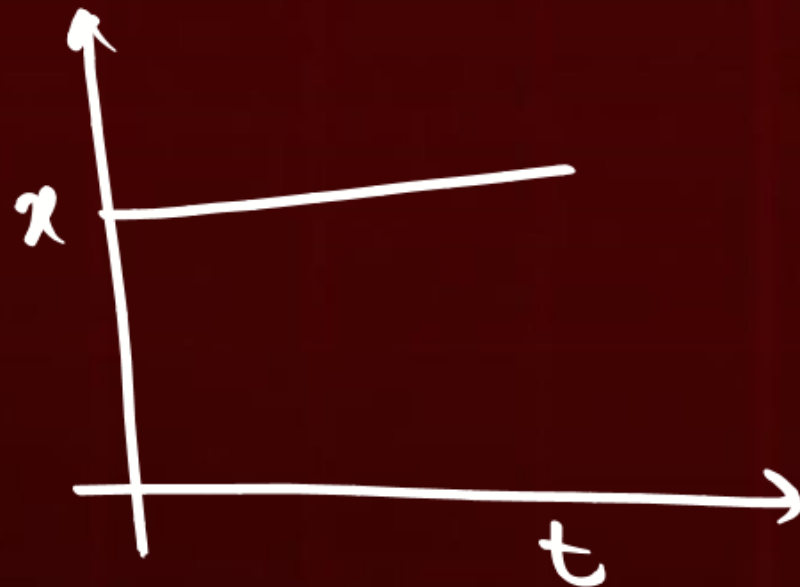
Q. 1



(a) uniform motion

(b) Non-uniform motion

Q. 2



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(a) velocity = c

(b) Body Rest

(c) $a=0$

(d) $a>0$

Thank
You